

The Effect of Note-Taking and Learning Styles to Comprehension and Memory Recall of the Students

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I. Introduction

Background of the Problem

Learning is a complex process that involves acquiring, understanding, and retaining information. Students differ in how they best absorb knowledge, and one key factor that influences this process is learning style. According to educational psychology, learners are often categorized as auditory, visual, or reading learners, each with a preferred mode of processing information. Auditory learners grasp concepts better through listening and discussion, visual learners benefit from diagrams, charts, and visual aids, while reading learners prefer written materials. However, regardless of learning style, students often rely on note-taking as a strategy to enhance comprehension and retention during lessons.

Note-taking is more than just writing down information; it is an active learning process that helps students encode and organize knowledge. Studies suggest that students who take notes tend to recall information better and perform more effectively in comprehension tasks compared to those who do not. Yet, the effectiveness of note-taking may vary depending on a learner's dominant learning style. For instance, a visual learner may benefit more from organized, diagram-based notes, while auditory learners might find less benefit from written notes alone. This creates a need to explore how note-taking interacts with learning styles in affecting students' comprehension and memory recall.

Given the increasing emphasis on independent and effective learning, understanding the relationship between note-taking and learning styles can help educators develop strategies that align with students' cognitive preferences. By examining this interaction, this study aims to provide insights into how different types of learners can maximize comprehension and memory performance through appropriate note-taking practices.

Importance of the Experiment

This experiment is significant because it will help determine whether note-taking truly enhances comprehension and memory recall across different learning styles. The findings can guide teachers in designing more personalized learning approaches and encourage students to adopt study habits suited to their strengths. Moreover, understanding these relationships can improve academic performance and promote more effective study techniques, especially in classrooms where diverse learning styles coexist.

Research Objectives:

- To determine the effect of note-taking on students' comprehension and memory recall.
- To identify which learning style; auditory, visual, or reading, shows the highest level of comprehension and memory recall.
- To examine the interaction effect between note-taking and learning styles on students' comprehension and memory recall.

Hypothesis to be Tested

Null Hypothesis (H₀): There is no significant effect of note-taking and learning styles on the comprehension and memory recall of students.

Alternative Hypothesis (H₁): There is a significant effect of note-taking and learning styles on the comprehension and memory recall of students.

II. Review of Related Literature

Literature Review: Sensory Modalities, Integration, and Comprehension in Reading and Note-Taking

Learning styles are often described in terms of preferred sensory modalities such as visual, auditory, and read/write, as in the VARK model. These styles of learning suggest that students may process and retain information differently when a material is presented as spoken text, written passages, or visual media. As early research on reading development emphasized the importance of integrating auditory and visual (AVI) information, Kahn and Birch (1968) reported that elementary school students with stronger auditory–visual integration skills tended to achieve higher reading scores even after controlling for IQ, while Evans (1969) argued that linking auditory patterns to printed symbols forms a sensory foundation for successful reading.

In a more recent study, researchers have highlighted cognitive mechanisms such as auditory–visual integration. Mossbridge et al. (2017) found that the ability to synchronize auditory and visual information—such as aligning phonological and orthographic cues—was associated with better reading comprehension, which suggests that modality integration is not merely sensory but also supports more precise timing and text processing. Earlier instructional studies further show that tailoring activities to specific modalities can influence comprehension. Kennedy and Weener (1973) observed that visual exercises improved both reading and listening comprehension, while auditory training led to greater improvements in listening alone, indicating that instructional effects are modality-specific. In higher education, Marlia and Si (2025) reported that kinesthetic reading techniques, such as read-write and read-type, produced greater comprehension improvements than purely visual or auditory approaches, consistent with levels-of-processing theory. Similarly, Herlita (2025) found that VARK-based instruction improved high school reading outcomes by accommodating diverse sensory preferences.

Debates about digital audiobooks also underline the importance of modality. Kosch et al. (2025) noted that some scholars regard listening to literature as equivalent to reading, whereas others argue that different sensory reception creates distinct aesthetic experiences. Together, these findings suggest that learning outcomes in reading tasks may depend on how information is delivered and how well it aligns with students' dominant or preferred modalities.

Note-taking is a widely used strategy for supporting learning during lectures and reading tasks. The memory benefits of note-taking are often described in terms of two functions which are external storage effect, in which written notes provide a

record for later review, and an encoding effect, in which the act of taking notes promotes deeper processing of information (Di Vesta & Gray, 1972, as cited in Jansen, Lakens, & IJsselsteijn, 2017). Siegel (2025) found characteristics of note-takers who took extensive, average, and few notes that were linked to variations in performance and opinions regarding the usefulness of notes.

In listening contexts, combining note-taking with metacognitive strategies can enhance comprehension. Ghorbannejad (2025) showed that teaching note-taking and metacognitive listening strategies led to significant gains in listening comprehension. Conversely, visual activities that compete for attention may interfere with memory. Ifroz et al. (2025) reported that doodling and drawing during an audio listening task impaired young adults' later recall, implying that not all forms of writing or visual engagement support learning. Obergassel et al. (2025) found that cognitive-load-adapted quizzing produced better conceptual retention than note-taking. These studies indicate that the effectiveness of note-taking depends on how it is implemented and what other demands are placed on learners' attention.

Overall, prior research shows that both sensory modality and note-taking practices influence how learners comprehend and remember information. However, most studies have examined these factors separately rather than in combination. To address this gap, the present study uses a 2×3 factorial design to investigate the separate and combined effects of learning styles (auditory, reading, visual) and note-taking (with notes, without notes) on students' comprehension and memory recall.

III. Methodology

Design: 2x3 Factorial Design

Factors and Levels:

Factor A: Learning Styles

Level 1: Auditory

Level 2: Reading

Level 3: Visual

Factor B: Presence of Note-Taking

Level 1: With Notes

Level 2: Without Notes

Response Variables: Comprehension (short quiz, 5 items) and Memory Recall (short quiz, 5 items)

Sample and Replication: 36 participants (12 per learning styles and 18 per presence of note-taking)

Randomization: Participants were randomly assigned to combinations of learning styles (auditory, reading, visual/video) and note-taking condition (with or without note-taking) by randomly picking from pre-coded shuffled consent forms.

Data Collection Procedure:

1. A 10-item multiple choice pre-test (about earthquake) was prepared. 5 of those questions are for comprehension, the other 5 are for testing the memory recall of the respondents. A 10-item multiple choice post-test (about the passage) was also prepared. 5 of those questions are for comprehension, and the other 5 are for memory recall of the respondents).
2. Each individual was given a consent form, which has a code to identify the note-taking strategies and learning styles of the respondents. The consent forms were shuffled before distribution so that assignment was effectively random. The respondents will be assigned to either listening/reading/visualizing a passage based on the identifier.
3. During the pre-test, each respondent was given 2 minutes to answer the questions about earthquakes.
4. After the pre-test, each respondent was given a passage for 2 minutes and 10 seconds based on the identifier from the consent form.
5. The 36 respondents were divided into three groups of 12 based on modality: auditory, reading, and visual/video. In each modality group, 6 respondents were allowed to take notes while the remaining 6 were not. In the auditory condition, all respondents used the same device with earphones and listened (audio is in full volume) to the passage. In the reading condition, all respondents were given identical printed copies of the passage. In the visual/video condition, all respondents used the same device to watch the video of the passage (audio is set to full volume and the screen brightness is set to maximum).
6. After the passage, the participants were given a post-test.
7. During the post-test, each participant was given 2 minutes to answer the questions based on the passage.
8. The pre-test and post-test scores were recorded.

IV. Results

Descriptive Statistics

Descriptive Statistics									
	N	Minimum	Maximum	Mean	Std. Deviation	Skewness		Kurtosis	
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error	Statistic	Std. Error
PostTestK	36	2	5	3.92	.806	-.190	.393	-.630	.768
PostTestU	36	1	5	3.97	1.276	-.994	.393	-.188	.768
Valid N (listwise)	36								

Figure 1 shows the descriptive statistics results that summarize the scores of the two dependent variables which are the comprehension (PostTestK) and memory recall (PostTestU). The comprehension scores ranged from 2 to 5, with an average score of 3.92 and a standard deviation of 0.806, indicating that most students' scores were relatively close to the mean. For memory recall, the scores ranged from 1 to 5, with a mean score of 3.97 and a standard deviation of 1.276, showing more variability in students' performance compared to the comprehension scores (PostTestK).

Assumption 5:

a. Test of Univariate Outliers

Descriptive Statistics									
	N	Minimum	Maximum	Mean	Std. Deviation	Skewness		Kurtosis	
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error	Statistic	Std. Error
PostTestK	36	2	5	3.92	.806	-.190	.393	-.630	.768
PostTestU	36	1	5	3.97	1.276	-.994	.393	-.188	.768
Valid N (listwise)	36								

Figure 2 shows the results of descriptive statistics for the two dependent variables; the comprehension (PostTestK) and memory recall (PostTestU). The skewness values for comprehension (-0.190) and memory recall (-0.994) fall within the acceptable range of -1 to +1, indicating that the scores revealed minimal skewness and no univariate outlier. This suggests that no participants had unusually extreme scores that could distort the group means or affect statistical assumptions. This means that the dataset is stable and suitable for multivariate analysis without requiring data removal or adjustment.

b. Test of Multivariate Outliers

Residuals Statistics ^a					
	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	16.07	22.85	18.50	1.945	36
Std. Predicted Value	-1.250	2.236	.000	1.000	36
Standard Error of Predicted Value	1.787	5.548	2.966	.838	36
Adjusted Predicted Value	14.44	23.96	18.73	2.324	36
Residual	-16.459	17.930	.000	10.355	36
Std. Residual	-1.543	1.681	.000	.971	36
Stud. Residual	-1.582	1.756	-.010	1.011	36
Deleted Residual	-17.296	19.565	-.234	11.236	36
Stud. Deleted Residual	-1.621	1.817	-.008	1.024	36
Mahal. Distance	.011	8.500	1.944	1.778	36
Cook's Distance	.000	.140	.029	.031	36
Centered Leverage Value	.000	.243	.056	.051	36

a. Dependent Variable: ParticipantNo

PValueMAH		
	.9946	
	.9946	
	.9946	
	.9946	
	.7122	.2448
	.7122	.2448
	.7122	.2431
	.7122	.2431
	.7122	.2391
	.6967	.2391
	.6967	.2391
	.4820	.2391
	.4820	.0615
	.4793	.0432
	.4793	.0143
	.3769	
	.3769	
	.3769	
	.3769	
	.3769	
	.3769	
	.3769	
	.3667	

Figure 3 shows the result of residuals statistics used for detecting multivariate outliers through Mahalanobis distance. The p-values indicate that none of the cases exceeded the critical cutoff associated with $p\text{-value} < 0.001$, with the minimum Mahalanobis p-value of 0.0143. This implies that all participants fell within the acceptable multivariate range and no extreme score combinations were present that could distort the covariance structure. As a result, the dataset meets the assumption regarding the multivariate outlier.

Assumption 6: Test of Normality

Tests of Normality							
		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	LearningStyle	Statistic	df	Sig.	Statistic	df	Sig.
PostTestK	Auditory	.257	12	.028	.807	12	.011
	Reading	.237	12	.061	.891	12	.123
	Visual	.279	12	.011	.784	12	.006
PostTestU	Auditory	.357	12	<.001	.710	12	.001
	Reading	.282	12	.009	.770	12	.004
	Visual	.238	12	.059	.798	12	.009

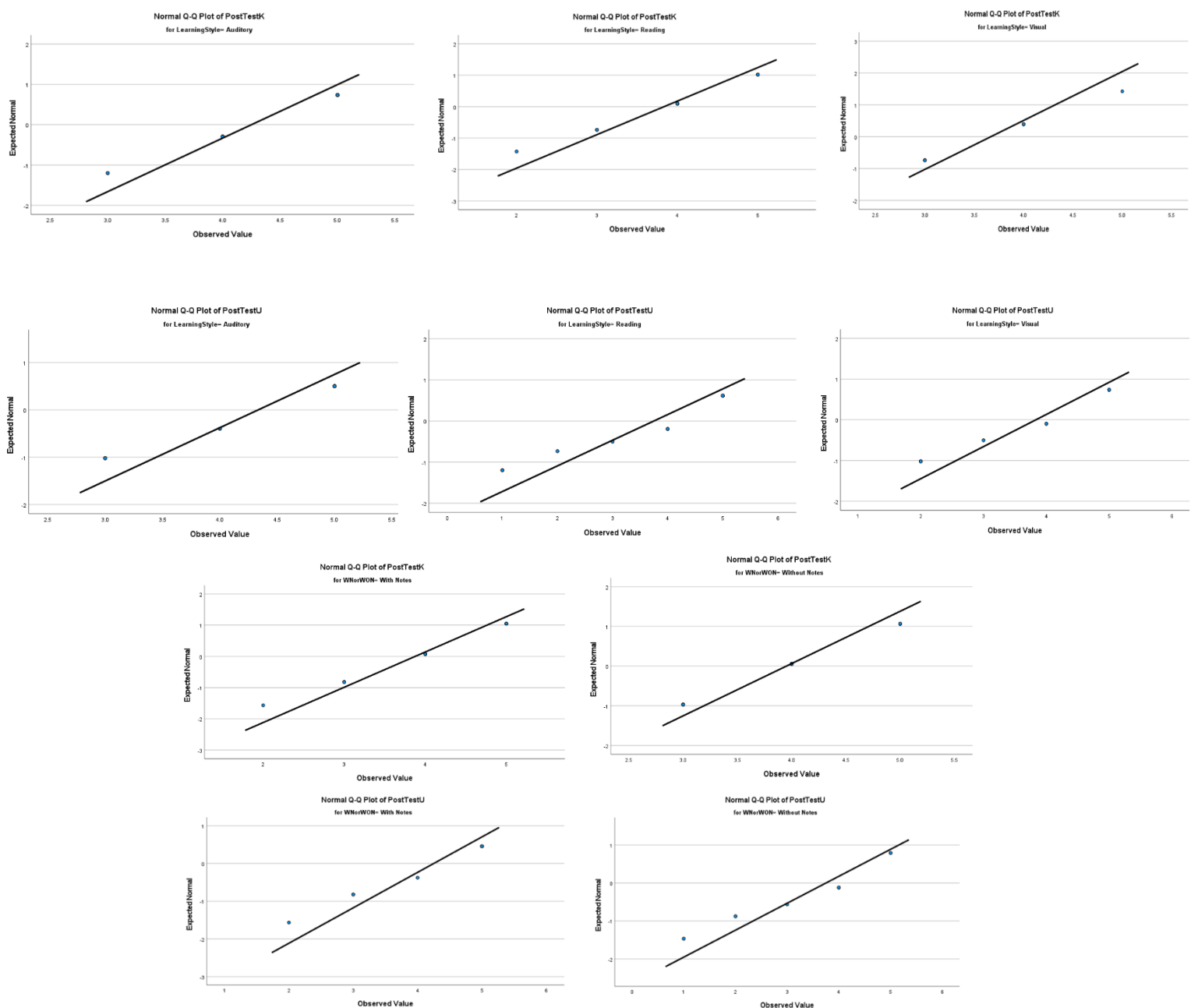
a. Lilliefors Significance Correction

Tests of Normality							
		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	WNorWON	Statistic	df	Sig.	Statistic	df	Sig.
PostTestK	With Notes	.244	16	.012	.878	16	.036
	Without Notes	.226	20	.009	.816	20	.002
PostTestU	With Notes	.384	16	<.001	.703	16	<.001
	Without Notes	.270	20	<.001	.807	20	.001

a. Lilliefors Significance Correction

Figure 4 and 5 show the Shapiro-Wilk normality test results for comprehension (PostTestK) and memory recall (PostTestU) across the levels of learning styles and the presence or absence of note-taking. Most groups showed $p\text{-value} < 0.05$, indicating violation of normality, except for the Reading group's comprehension scores in Figure 4, which met the normality criterion ($p = 0.123 > 0.05$). A similar pattern of non-normality was evident in Figure 5, where both comprehension and memory recall scores for the With Notes and Without Notes groups were less than the 0.05 level of significance. Despite these normality violations, MANOVA and related ANOVA analyses are generally robust to moderate deviations from normality, especially with balanced group sizes. Therefore, proceeding with MANOVA remains acceptable without requiring data transformation and non-parametric alternatives.

Assumption 7: Test of Linearity



Figures 6 to 15 show the Normal QQ plots used to assess the behavior of the two dependent variable scores. The points mostly followed the diagonal lines with only small deviations at the tails, which indicates approximate normality. Since the QQ plots show that the variables were roughly distributed, it is reasonable to assume that a linear relationship exists between the comprehension (PostTestK) and memory recall scores (PostTestU) for each group of the independent variables. This suggests that the linearity assumption required for MANOVA is met.

Assumption 8:

a. Test of Homogeneity of Variance

Levene's Test of Equality of Error Variances^a

		Levene Statistic	df1	df2	Sig.
PostTestK	Based on Mean	.990	5	30	.441
	Based on Median	.943	5	30	.468
	Based on Median and with adjusted df	.943	5	19.896	.475
	Based on trimmed mean	.994	5	30	.438
PostTestU	Based on Mean	2.174	5	30	.084
	Based on Median	.725	5	30	.610
	Based on Median and with adjusted df	.725	5	24.797	.611
	Based on trimmed mean	2.018	5	30	.105

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Design: Intercept + LearningStyle + WNorWON + LearningStyle * WNorWON

Figure 16 shows the Levene's Test of Equality of Error Variance results for both comprehension (PostTestK) and memory recall (PostTestU) scores. All p-values are greater than 0.05, indicating that the variances among the groups are not significantly different. This shows that the data meet the assumption of homogeneity of variance, as all the p-values exceed the required cutoff.

b. Test of Homogeneity of Covariance

Box's Test of Equality of Covariance Matrices^a

Box's M	12.335
F	.670
df1	15
df2	2618.490
Sig.	.816

Tests the null hypothesis that the observed covariance matrices of the dependent variables are equal across groups.

a. Design: Intercept + LearningStyle + WNorWON + LearningStyle * WNorWON

Figure 17 shows the Box's Test of Equality of Covariance Matrices results of the comprehension (PostTestK) and memory recall (PostTestU) across all groups of learning styles and presence or absence of note-taking. The test revealed a Box's M value of 12.335, with F-value = 0.670 and p-value = 0.816. Since the p-value is greater than the 0.05 level of significance, there is no significant difference in the covariance matrices of the dependent variables across the independent variable groups. This means that the relationship between comprehension and memory recall

remains consistent regardless of learning style or note-taking condition. This confirms the assumption of homogeneity of covariance matrices which is important for multivariate comparisons.

Assumption 9: Test of Multicollinearity

Correlations			
		PostTestK	PostTestU
PostTestK	Pearson Correlation	1	.331 [*]
	Sig. (2-tailed)		.049
	N	36	36
PostTestU	Pearson Correlation	.331 [*]	1
	Sig. (2-tailed)	.049	
	N	36	36

*. Correlation is significant at the 0.05 level (2-tailed).

Figure 18 shows the correlation between comprehension (PostTestK) and memory recall (PostTestU) scores. The Pearson correlation was low ($r = .331$), showing only a weak relationship between the two measures. This value falls within the acceptable range of 0.10 to 0.90, which means it does not reach the cutoff point that would indicate multicollinearity. Therefore, both variables can be used together in the analysis without causing any issues in the results.

Two-way MANOVA Result:

a. Multivariate Test

Multivariate Tests ^a								
Effect		Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared	Noncent. Parameter
Intercept	Pillai's Trace	.968	432.749 ^b	2.000	29.000	<.001	.968	865.499
	Wilks' Lambda	.032	432.749 ^b	2.000	29.000	<.001	.968	865.499
	Hotelling's Trace	29.845	432.749 ^b	2.000	29.000	<.001	.968	865.499
	Roy's Largest Root	29.845	432.749 ^b	2.000	29.000	<.001	.968	865.499
LearningStyle	Pillai's Trace	.114	.903	4.000	60.000	.468	.057	3.613
	Wilks' Lambda	.886	.901 ^b	4.000	58.000	.469	.059	3.604
	Hotelling's Trace	.128	.897	4.000	56.000	.472	.060	3.587
	Roy's Largest Root	.128	1.919 ^c	2.000	30.000	.164	.113	3.839
WNorWON	Pillai's Trace	.062	.966 ^b	2.000	29.000	.392	.062	1.932
	Wilks' Lambda	.938	.966 ^b	2.000	29.000	.392	.062	1.932
	Hotelling's Trace	.067	.966 ^b	2.000	29.000	.392	.062	1.932
	Roy's Largest Root	.067	.966 ^b	2.000	29.000	.392	.062	1.932
LearningStyle * WNorWON	Pillai's Trace	.160	1.300	4.000	60.000	.280	.080	5.201
	Wilks' Lambda	.841	1.307 ^b	4.000	58.000	.278	.083	5.227
	Hotelling's Trace	.187	1.310	4.000	56.000	.278	.086	5.240
	Roy's Largest Root	.180	2.705 ^c	2.000	30.000	.083	.153	5.409

a. Design: Intercept * LearningStyle * WNorWON * LearningStyle * WNorWON

b. Exact statistic

c. The statistic is an upper bound on F that yields a lower bound on the significance level.

d. Computed using alpha = .05

Figure 19 shows the Two-way MANOVA results using Wilks' Lambda as the primary basis for interpretation. As shown, the multivariate test for learning style produced a Wilks' Lambda = 0.886, $F(4, 58) = 0.901$, $P\text{-value} = 0.469 > 0.05$ and partial eta square = 0.059 which denotes very small effect size, indicating that learning style does not have a statistically significant effect on the combined dependent variables of comprehension and memory recall. Similarly, the note-taking factor produced a Wilks' Lambda = .938 $F(2, 29) = 0.966$, $P\text{-value} = 0.392 > 0.05$ and partial eta square = 0.062 which also denotes a very small effect size, showing that

the presence or absence of note-taking does not significantly influence the multivariate outcomes.

b. Test of Between-Subject Effects

Source	Dependent Variable	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	PostTestK	3.917 ^a	5	.783	1.248	.312	.172
	PostTestU	6.097 ^b	5	1.219	.719	.614	.107
Intercept	PostTestK	522.667	1	522.667	832.566	<.001	.965
	PostTestU	562.602	1	562.602	331.755	<.001	.917
LearningStyle	PostTestK	2.362	2	1.181	1.881	.170	.111
	PostTestU	1.657	2	.829	.489	.618	.032
WNorWON	PostTestK	.107	1	.107	.170	.683	.006
	PostTestU	2.042	1	2.042	1.204	.281	.039
LearningStyle * WNorWON	PostTestK	1.670	2	.835	1.330	.280	.081
	PostTestU	1.862	2	.931	.549	.583	.035
Error	PostTestK	18.833	30	.628			
	PostTestU	50.875	30	1.696			
Total	PostTestK	575.000	36				
	PostTestU	625.000	36				
Corrected Total	PostTestK	22.750	35				
	PostTestU	56.972	35				

a. R Squared = .172 (Adjusted R Squared = .034)

b. R Squared = .107 (Adjusted R Squared = -.042)

Figure 20 shows the test of between-subjects effects results from the Two-Way MANOVA assessing the effects of learning styles, note-taking and their interaction on each dependent variable separately. Consistent with the multivariate findings, none of the univariate tests reached statistical significance. For learning style, the test revealed $F(2,30) = 1.861$, $p = 0.170 > 0.05$, partial eta squared = 0.111 for comprehension (PostTestK) and $F(2,30) = 0.489$, $p = 0.618 > 0.05$, partial eta squared = 0.032 for memory recall (PostTestU). Based on the study criteria ($0.06-0.13 = \text{medium}$), the effect of learning style on comprehension (PostTestK) reflects a medium effect size, whereas the effect on memory recall (PostTestU) shows small effect size. However, despite such effect size, both remain statistically insignificant.

The results also show that note-taking (W/ or W/O N) did not have a statistically significant effect on either comprehension or memory recall. For comprehension (PostTestK), the analysis reported $F(1,30) = 0.170$, $p = 0.683 > 0.05$, with a very small partial eta squared of 0.006, indicating almost no effect. Similarly, for memory recall (PostTestU), the results showed $F(1,30) = 1.204$, $p = 0.281 > 0.05$, with partial eta squared = 0.039, suggesting that note-taking alone did not meaningfully influence students' performance. These findings imply that whether students took notes or not did not significantly change their test outcomes in this particular sample.

Overall, these results suggest that neither learning style nor note-taking significantly influenced comprehension or memory recall in this study. Although some effect sizes (partial eta squared) show small to moderate values, especially for learning style, these were not large enough to reach statistical significance. This implies that, within this group of participants, students performed similarly on the post-tests regardless of their learning style category or whether they took notes. The findings may point to the possibility that other factors not measured in the study such as prior knowledge, study habits, motivation, or instructional quality could have had a stronger influence on student performance than the variables examined.

V. Discussion

The findings of this experiment were evaluated in relation to its objectives: to determine the effect of note-taking on students' comprehension and memory recall, to identify which learning style; auditory, visual, or reading demonstrates the highest performance, and to examine the interaction between note-taking and learning styles.

For the assumption checks, the dataset behaved well enough for Two-Way MANOVA. No univariate and multivariate outliers were detected based on the skewness values (skewness for comprehension = - 0.190; for recall = - 0.994, both within the acceptable range of -1 and +1) and Mahalanobis distance p-values above 0.001 cutoff. Normality showed minor deviations but remained acceptable for MANOVA, which is known to be robust to slight non-normality. Linearity among the dependent variables was observed in Normal QQ Plots. Homogeneity of variance was met, with all Levene's test p-values greater than 0.05, and homogeneity of covariance matrices was also satisfied (Box's M = 0.816). The two dependent variables were weakly correlated ($r = 0.33$, $p\text{-value} = 0.49$) which ruled out multicollinearity issues. Therefore, the assumption tests supported the use of a two-way MANOVA.

For the multivariate analysis, no significant differences were found in comprehension and memory recall across note-taking conditions or learning styles. Furthermore, the interaction effect between note-taking and learning styles did not yield statistically significant results. These outcomes suggest that, within the scope of this study, neither note-taking nor learning style had a measurable impact on students' post-test performance.

Based on the findings, the experiment failed to reject the null hypothesis, which states that note-taking and learning styles have no significant effect on comprehension and memory recall. The alternative hypothesis was not supported, as statistical tests failed to demonstrate meaningful differences or interactions among the variables studied.

Several factors may account for the lack of significant findings. First, the uniformity in mean scores across groups could indicate limited variability in performance. Second, the study did not control for the quality or method of note-taking, which may have influenced its effectiveness. Third, the categorization of learning styles may have oversimplified individual differences in cognitive processing, as students often exhibit mixed or situational preferences.

Lastly, the comprehension and recall tasks may not have been sufficiently tailored to highlight the strengths of each learning style, reducing the likelihood of detecting significant effects. This experiment also faced several limitations. The sample size of 36 participants may have limited the statistical power needed to detect subtle effects. Potential biases, such as prior exposure to note-taking strategies or self-perceived learning styles, may have influenced participant behavior. Additionally, uncontrolled variables such as motivation, prior knowledge, and environmental distractions were not accounted for and may have impacted the outcomes.

VI. Conclusion & Recommendations

This experiment explored the effects of note-taking practices and learning styles on students' comprehension and memory recall, and the results showed no significant differences across note-taking conditions, learning style groups, or their interaction. These findings indicate that students performed similarly regardless of how they took notes or which learning style they were assigned, suggesting that other factors beyond these variables may have played a larger role in their post-test performance. Although the study had limitations such as a small sample size and several uncontrolled factors, it still shows that commonly assumed strategies like following a preferred learning style or depending on note-taking do not automatically lead to better learning outcomes.

Based on these findings, the following recommendations are suggested for future research. Future research should include larger sample sizes and improved control of statistical assumptions to increase the reliability of the results. Researchers should also refine learning style classifications and ensure that note-taking methods are standardized to better capture their true effects. Lastly, future studies should minimize uncontrolled factors such as prior knowledge, motivation, and environmental distractions to obtain more reliable results, so they can better understand the relationship between note-taking, learning styles, and academic performance.

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VIII. Appendix

Raw Data

Participant No.	Materials Used	Learning Style	Presence of Note-Taking	Overall Score (Pre-Test - Knowledge and Understanding)	Score (Post-Test - Knowledge)	Score (Post-Test - Understanding)
FV001	Laptop (<i>Lenovo</i>) Yellow Paper (<i>Valiant</i>)	3	1	10	3	2
FR002	Short Bond Paper (<i>Arial</i> , 11)	2	2	9	5	4
FR003	Short Bond Paper (<i>Arial</i> , 11), Yellow Paper (<i>Valiant</i>)	2	1	9	3	3
FV004	Laptop (<i>Lenovo</i>)	3	2	10	5	5
FR005	Short Bond Paper (<i>Arial</i> , 11)	2	2	10	3	5
FA006	Cellphones (<i>Realme</i>), Earphones (<i>Miniso</i>), Yellow Paper (<i>Valiant</i>)	1	1	9	4	3
MR007	Short Bond Paper (<i>Arial</i> , 11)	2	2	9	5	5
MR008	Short Bond Paper (<i>Arial</i> , 11)	2	2	9	4	2

MV009	Laptop (<i>Lenovo</i>), Yellow Paper (<i>Valiant</i>)	3	1	9	3	3
FR010	Short Bond Paper (<i>Arial, 11</i>)	2	2	9	4	5
FA011	Cellphones (<i>Xiaomi</i>), Earphones (<i>UGreen</i>), Yellow Paper (<i>Valiant</i>)	1	1	10	5	5
MV012	Laptop (<i>Lenovo</i>)	3	2	10	3	2
MR013	Short Bond Paper (<i>Arial, 11</i>)	2	2	10	3	1
MA014	Cellphones (<i>Realme</i>), Earphones (<i>Miniso</i>), Yellow Paper (<i>Valiant</i>)	1	1	10	4	5
MV015	Laptop (<i>Lenovo</i>), Yellow Paper (<i>Valiant</i>)	3	1	10	4	5
MA016	Cellphones (<i>Samsung</i>), Earphones (<i>SM Stationery</i>)	1	2	10	3	5
FV017	Laptop (<i>Lenovo</i>), Yellow Paper (<i>Valiant</i>)	3	1	10	4	5
FV018	Laptop (<i>Lenovo</i>), Yellow Paper (<i>Valiant</i>)	3	1	10	3	5
FR019	Short Bond Paper (<i>Arial, 11</i>), Yellow Paper (<i>Valiant</i>)	2	1	8	2	5
FA020	Cellphones (<i>Xiaomi</i>), Earphones (<i>UGreen</i>)	1	2	9	5	5
MV021	Laptop (<i>Lenovo</i>)	3	2	10	4	4
MV022	Laptop (<i>Lenovo</i>), Yellow Paper (<i>Valiant</i>)	3	1	10	4	4
FA023	Cellphones (<i>Samsung</i>), Earphones (<i>SM Stationery</i>), Yellow Paper (<i>Valiant</i>)	1	1	10	5	5
FA024	Cellphones (<i>Realme</i>), Earphones (<i>Miniso</i>)	1	2	10	4	3
MV025	Laptop (<i>Lenovo</i>)	3	2	10	3	4
MR026	Short Bond Paper (<i>Arial, 11</i>), Yellow Paper (<i>Valiant</i>)	2	1	10	4	5
FR027	Short Bond Paper (<i>Arial, 11</i>)	2	2	9	4	4

FA028	Cellphones (<i>Samsung</i>), Earphones (<i>SM Stationery</i>), Yellow Paper (<i>Valiant</i>)	1	1	10	5	5
MR029	Short Bond Paper (<i>Arial, 11</i>)	2	2	9	4	1
MA030	Cellphones (<i>Samsung</i>), Earphones (<i>SM Stationery</i>)	1	2	10	4	4
FA031	Cellphones (<i>Xiaomi</i>), Earphones (<i>UGreen</i>), Yellow Paper (<i>Valiant</i>)	1	1	10	4	3
FR032	Short Bond Paper (<i>Arial, 11</i>), Yellow Paper (<i>Valiant</i>)	2	1	9	5	5
FA033	Cellphones (<i>Realme</i>), Earphones (<i>Miniso</i>)	1	2	9	3	4
FA034	Cellphones (<i>Xiaomi</i>), Earphones (<i>UGreen</i>)	1	2	9	5	5
FV035	Laptop (<i>Lenovo</i>)	3	2	9	4	5
FV036	Laptop (<i>Lenovo</i>)	3	2	9	4	2

LEGEND:

Learning Style:

Auditory - 1

Reading - 2

Visual - 3

Presence of Note-Taking:

With Notes - 1

Without Notes - 2

Consent Form

Participant No. _____

Participant Consent Form

You are invited to participate in an experimental study titled ***"The Effect of Note-Taking and Learning Styles to Comprehension and Memory Recall of the Students"*** conducted by ***"Data the Explorer"*** group. This experiment aims ***to investigate how different note-taking strategies and individual learning styles influence students' ability to understand and remember information.***

If you agree to participate, you will be tasked to:

- Answer the pre-test knowledge questions.
- You will be assigned to either listening/ reading/ visualizing a passage for 2 minutes.
- Answer the post-test knowledge questions.

Participation is entirely voluntary. You may refuse or withdraw at any time without loss of benefits.

Your responses will be kept confidential. Data will be anonymized using your participant number and stored securely.

By checking below, you confirm that:

- You have read and understood the information above.
- You voluntarily agree to participate.
- You understand you may withdraw at any time.